Compute-Efficient Geometric Adaptation of Stable Virtual Camera for Trajectory Novel View Synthesis

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Abstract

This project studies geometric adaptation of Stable Virtual Camera (SEVA) for trajectory novel view synthesis. The goal is to improve view-to-view geometric consistency using an epipolar loss based on known source and target cameras. To achieve this, this project evaluates two compute-aware strategies: low-noise epipolar fine-tuning and pose-adaptive inference sampling. Preliminary results on a single scene `000002` *show that epipolar loss is more effective when restricted to low-noise timesteps, while pose-adaptive sampling improves perceptual quality without any model update. These findings suggest that timestep-aware geometric supervision and sampling design are practical directions for improving SEVA under limited compute.*

1. Introduction

Stable Virtual Camera (SEVA) [2] is a pretrained diffusion model for camera-controlled novel view synthesis. Given one or more input images and target camera poses, SEVA can synthesize novel views or full camera trajectories. However, even when individual generated frames look visually plausible, trajectory outputs can still suffer from geometric inconsistency, including texture drift, unstable view-to-view correspondence, foreground-background detachment, and inconsistent object boundaries across frames.

The goal of this project is to improve this geometric consistency during fine-tuning. Because source and target camera poses are known during training, epipolar geometry provides a natural supervision signal: a target pixel should correspond to some point along the correct epipolar line in an input view. This motivates an epipolar loss that encourages generated target views to remain geometrically consistent with source views, without requiring depth supervision or changes to the SEVA architecture.

A key challenge is that SEVA is a large pretrained model, so full-backbone fine-tuning is expensive under a limited compute budget. This makes it difficult for a weak auxiliary geometric loss to noticeably change model behavior unless it is applied in the right regime. In particular, an epipolar loss computed from decoded predicted clean images may be unreliable at high diffusion noise levels, where the predicted image is still far from the final output. For this reason, this project evaluates low-noise epipolar fine-tuning, where the geometric loss is applied when the predicted clean image is closer to the final image manifold.

In addition to loss-based fine-tuning, this project explores an inference-time alternative: pose-adaptive sampling. SEVA's trajectory inference uses a two-pass sampling procedure in which anchor frames are generated first and intermediate frames are generated between anchors. The default interpolation strategy selects anchors mostly according to frame index, but frame index does not always reflect camera-motion difficulty. This project instead selects anchors according to cumulative camera motion, placing more anchors in regions with larger translation or rotation while enforcing the context-window constraint required by SEVA's second-pass sampler.

The preliminary experiments are conducted on a single scene, with 349 evaluated target frames. The results show that epipolar loss is more effective when restricted to low-noise timesteps, where the predicted clean images are more meaningful for geometric supervision. The results also show that pose-adaptive sampling improves perceptual quality without changing the model weights, suggesting that sampling design is a practical low-compute direction for improving SEVA trajectory synthesis.

The contributions of this individual project are:

- A visibility-gated epipolar distribution loss for geometry-aware fine-tuning without depth supervision or architecture changes.
- A low-noise timestep analysis showing that epipolar supervision is more effective when applied near the clean-image regime.
- A pose-adaptive anchor sampler for trajectory inference that selects anchors by camera-motion arc length and improves perceptual quality without model fine-tuning.
- An empirical comparison between loss-based fine-tuning and sampling-based adaptation under a limited compute setting.

2. Related Work

Novel view synthesis. Classical NVS methods synthesize unseen viewpoints using explicit geometry, multi-view reconstruction, or scene-specific representations. Neural radiance fields and 3D Gaussian representations provide high-quality scene-specific rendering but typically require optimization or reconstruction for each scene. Feed-forward NVS and generative NVS aim to reduce this per-scene cost by learning priors across many scenes.

Diffusion models for view synthesis. Diffusion models have recently been adapted to camera-controlled image and video generation. SEVA [2] is a generalist diffusion model for set and trajectory NVS, combining image conditioning, camera conditioning, and procedural sampling. This project does not introduce a new architecture. Instead, it studies practical adaptation of this pretrained model under limited compute.

Geometry-aware supervision. Epipolar constraints are a classical tool for multi-view geometry. Recent neural and generative methods use epipolar structure for matching, attention, or photometric consistency. The epipolar loss in this project differs from hard correspondence or depth-warp losses: it samples a set of possible source correspondences along the epipolar line and matches soft distributions. This avoids requiring depth and allows ambiguous or occluded pixels to be gated out.

Sampling for long trajectories. Long video generation often relies on anchors, keyframes, chunks, or memory. SEVA uses procedural sampling for long trajectories: anchors are generated first, and intermediate target frames are generated in chunks between anchors [2]. This project modifies the anchor-selection rule itself, choosing anchors according to camera-motion difficulty rather than uniform frame index.

3. Background: Stable Virtual Camera

Task formulation. Let the input views be

$$\mathcal{I}_{\text{in}} = \{(I_i, \mathbf{K}_i, \mathbf{T}_i)\}_{i=1}^{P}, \tag{1}$$

where I_i is an RGB image, $\mathbf{K}_i$ is the intrinsic matrix, and $\mathbf{T}_i$ is the camera-to-world pose. Given target cameras $\{(\mathbf{K}_j, \mathbf{T}_j)\}_{j=1}^{Q}$, the model predicts target images $\{\hat{I}_j\}_{j=1}^{Q}$.

SEVA conditioning. At a high level, SEVA conditions generation using image features, input-view VAE latents, camera information such as Plücker ray embeddings, and an input-frame mask. During training, input frames are represented using clean latent replacements, while target frames are noised according to the diffusion schedule. This distinction is important: if the input-view clean latent replacement is omitted during fine-tuning, the training setup no longer matches inference.

Two-pass trajectory inference. For long trajectories, all target frames cannot fit into a single context window. SEVA's trajectory inference therefore uses a two-pass procedure. The first pass generates anchor views. The second pass synthesizes intermediate frames in chunks bounded by neighboring anchors:

$$\{A_m, T_{m,1}, \ldots, T_{m,k}, A_{m+1}\}. \tag{2}$$

If the second-pass context length is T, then each chunk can contain at most $T - 2$ intermediate target frames because two slots are reserved for anchors. This constraint becomes central to the pose-adaptive sampler.

4. Experimental Setup

4.1. Task and data

The experiments use parsed trajectory scenes evaluated with the `img2vid` setting. Each scene contains posed RGB frames with known intrinsics and camera-to-world poses. The model receives six input views and generates a target camera trajectory.

The current results are reported on scene `000002`, with 349 evaluated target frames. This is a preliminary single-scene evaluation intended to compare design choices before scaling to a larger multi-scene benchmark.

4.2. Evaluation protocol

All methods are evaluated with the same image-level metrics:

$$\text{PSNR} \uparrow, \quad \text{SSIM} \uparrow, \quad \text{LPIPS} \downarrow. \tag{3}$$

These metrics measure frame-level reconstruction and perceptual similarity. They do not fully capture geometric trajectory consistency, so future evaluation should include epipolar or multi-view consistency metrics.

Unless otherwise stated, inference uses `img2vid` with six input views, `replace_or_include_input=True`, and `chunk_strategy=interp`. For the sampling ablation, the original SEVA weights are used and only the anchor selection strategy is changed.

5. Method

5.1. SEVA-compatible fine-tuning

The pretrained SEVA backbone is fine-tuned while keeping the VAE and image conditioner frozen. For a clean latent z_0, Gaussian noise $\boldsymbol{\epsilon}$, and noise level σ, the noised latent is

$$z_\sigma = z_0 + \sigma\boldsymbol{\epsilon}. \tag{4}$$

The model predicts noise, and the diffusion objective is computed on target frames:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}\left[\|\boldsymbol{\epsilon}_\theta(z_\sigma, c, t) - \boldsymbol{\epsilon}\|_2^2\right], \tag{5}$$

where c denotes image and camera conditioning. Input frames are replaced with their clean encoded latents before the U-Net forward pass, matching SEVA's inference behavior.

5.2. Visibility-gated epipolar distribution loss

For a sampled target pixel p_t, the corresponding source-view point must lie on the epipolar line

$$\ell_s = F_{s\leftarrow t} p_t, \tag{6}$$

where $F_{s\leftarrow t}$ is computed from the source and target intrinsics and camera poses. Because depth is unknown, the implementation samples K candidate points along the valid segment of this line:

$$\{q_s^k\}_{k=1}^K. \tag{7}$$

Fixed descriptors $\phi(\cdot)$ are extracted using RGB values and Sobel gradients. The ground-truth target feature defines a teacher distribution over line candidates:

$$p_{\text{gt}}(k) = \operatorname{softmax}\left(\frac{\operatorname{sim}\left(\phi(I_t, p_t), \phi(I_s, q_s^k)\right)}{\tau}\right). \tag{8}$$

The predicted target feature defines a student distribution:

$$p_{\text{pred}}(k) = \operatorname{softmax}\left(\frac{\operatorname{sim}\left(\phi(\hat{I}_t, p_t), \phi(I_s, q_s^k)\right)}{\tau}\right). \tag{9}$$

The epipolar loss is:

$$\mathcal{L}_{\text{epi}} = w_{\text{conf}}\, w_\sigma\, \mathrm{KL}\left(p_{\text{gt}} \| p_{\text{pred}}\right), \tag{10}$$

where w_{conf} suppresses ambiguous or occluded pixels and w_σ downweights high-noise timesteps. The full training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda_{\text{epi}}\mathcal{L}_{\text{epi}}. \tag{11}$$

5.3. Low-noise timestep restriction

The epipolar loss depends on decoded predicted clean images. At high diffusion noise levels, these predictions can be visually unreliable, causing epipolar distributions to become noisy or misleading. The experiments therefore evaluate restricted fine-tuning regimes where the maximum diffusion noise index is set to 50 or 100, focusing training on near-clean timesteps.

5.4. Pose-adaptive anchor sampling

The default `interp` sampler selects anchor frames approximately uniformly in frame index. However, frame index does not necessarily reflect camera difficulty. This project defines a camera-motion step between consecutive frames:

$$d_i = \sqrt{\alpha\left(\frac{\|\mathbf{C}_{i+1} - \mathbf{C}_i\|}{s_C}\right)^2 + \beta\left(\frac{\theta(\mathbf{R}_i^\top \mathbf{R}_{i+1})}{s_R}\right)^2}, \tag{12}$$

where $\mathbf{C}_i$ and $\mathbf{R}_i$ are the camera center and rotation, θ is the full 3D relative rotation angle, and s_C, s_R are normalization constants. Anchors are selected uniformly in cumulative camera-motion arc length:

$$D_j = \sum_{i<j} d_i. \tag{13}$$

Context-window repair. Because each second-pass chunk can contain at most $T-2$ target frames, pose-adaptive anchor selection may leave long low-motion intervals that are invalid for SEVA's chunking rule. The implementation therefore inserts additional anchors until the raw frame-index gap satisfies:

$$a_{m+1} - a_m \le T - 2. \tag{14}$$

This preserves SEVA's two-pass sampler while making anchor placement sensitive to camera motion.

6. Experiments

6.1. Setup

The experiments are conducted on scene `000002`, with 349 evaluated target frames. All runs use the `img2vid` evaluation path with six input views. Unless otherwise stated, frame-averaged PSNR, SSIM, and LPIPS are reported over the generated target trajectory. These results are preliminary and should be interpreted as a single-scene individual-project ablation rather than a final multi-scene benchmark.

The experiments are separated into two groups. First, inference-time sampling is evaluated using the original SEVA weights. Second, fine-tuning experiments compare the original diffusion loss against the epipolar loss under matched low-noise timestep settings. This separation is important because sampling changes do not update model weights, while fine-tuning results depend heavily on optimization budget and learning rate.

6.2. Inference-Time Sampling Ablation

The first experiment tests whether pose-adaptive anchor selection improves trajectory synthesis without any fine-tuning. Both rows in Table 1 use the original SEVA checkpoint. The only difference is the anchor selection strategy used by the two-pass `interp` sampler.

Sampling	PSNR↑	SSIM↑	LPIPS↓
Uniform interp	14.170	**0.550**	0.382
Pose-adaptive interp	**14.313**	0.544	**0.374**

Table 1. Inference-time sampling ablation using the original SEVA weights on scene `000002`. Pose-adaptive sampling improves PSNR and LPIPS without any model fine-tuning, although SSIM decreases slightly.

Loss	Max timestep	PSNR↑	SSIM↑	LPIPS↓
Original	50	14.128	0.543	0.394
Epipolar	50	**14.352**	**0.559**	**0.383**
Original	100	14.043	0.543	0.393
Epipolar	100	**14.286**	**0.556**	**0.386**

Table 2. Controlled low-noise fine-tuning ablation on scene `000002`. For both timestep ranges, epipolar-loss fine-tuning improves over the corresponding original-loss fine-tuning baseline.

Pose-adaptive sampling improves PSNR from 14.170 to 14.313 and improves LPIPS from 0.382 to 0.374. SSIM decreases slightly from 0.550 to 0.544. Since this modification requires no gradient updates, this result suggests that camera-motion-aware anchor placement is a practical low-compute lever for trajectory synthesis. For an individual-project setting with limited GPU budget, this is an important observation: improving the inference sampler can provide measurable gains without additional fine-tuning cost.

6.3. Controlled Low-Noise Fine-Tuning Ablation

The second experiment compares original diffusion-loss fine-tuning with epipolar-loss fine-tuning under controlled low-noise timestep settings. The maximum sampled timestep is restricted to either 50 or 100, so the model is fine-tuned primarily near the clean-image regime. This tests the hypothesis that epipolar supervision is more meaningful when the decoded predicted clean image is not dominated by high diffusion noise.

The results in Table 2 show that epipolar supervision is more effective when restricted to low-noise timesteps. At max timestep 50, epipolar fine-tuning improves PSNR by 0.224, SSIM by 0.016, and LPIPS by 0.011 relative to the original-loss baseline. At max timestep 100, epipolar fine-tuning improves PSNR by 0.243, SSIM by 0.013, and LPIPS by 0.007. This supports the intuition that image-space geometric supervision should be applied when predicted clean images are visually meaningful.

6.4. Learning-Rate Stress Test

A high learning rate of 10^{-4} is also tested under max timestep 100. This setting is not treated as a competitive result; instead, it probes the stability of full-backbone SEVA fine-tuning.

Loss	LR	PSNR↑	SSIM↑	LPIPS↓
Original	10^{-4}	10.945	0.412	0.607
Epipolar	10^{-4}	11.516	0.425	0.581

Table 3. Learning-rate stress test with max timestep 100. A high learning rate substantially degrades both original-loss and epipolar-loss fine-tuning, indicating that the large pretrained SEVA backbone is sensitive to aggressive updates.

Both objectives degrade severely at 10^{-4}, with LPIPS increasing above 0.58. Although the epipolar-loss run remains slightly better than the original-loss run in this unstable regime, both are far worse than the stable low-noise runs. This result is therefore used only as evidence of learning-rate sensitivity.

6.5. Uncontrolled Full-Timestep Reference

Full-timestep fine-tuning runs were also performed, but they used a substantially different optimization budget. In particular, the max-timestep-1000 runs were trained with approximately $10\times$ more optimization steps than the low-noise runs. For this reason, they are excluded from the controlled comparison tables above.

As an uncontrolled reference, full-timestep original-loss fine-tuning reached 14.485 PSNR, 0.561 SSIM, and 0.382 LPIPS, while full-timestep epipolar-loss fine-tuning reached 14.301 PSNR, 0.556 SSIM, and 0.391 LPIPS. These numbers suggest that epipolar loss is not beneficial when applied across the full timestep range under that training configuration. However, because the training budget differs, these results should not be directly compared against the controlled low-noise experiments.

6.6. Summary of Findings

The current ablations suggest three main findings. First, pose-adaptive sampling improves perceptual quality without requiring any additional training, making it the most compute-efficient modification tested so far. Second, epipolar loss is timestep-sensitive: it improves over original-loss fine-tuning when the timestep range and training budget are matched in low-noise regimes, but it does not improve in the uncontrolled full-timestep setting. Third, full-backbone SEVA fine-tuning is sensitive to learning rate, as 10^{-4} causes severe degradation for both original and epipolar objectives.

These findings motivate the next stage of evaluation. The current metrics are image-level and framewise; they do not fully measure whether the generated trajectory follows the requested camera geometry. Future experiments should therefore add geometric consistency metrics such as symmetric epipolar distance, Sampson distance, or thresholded epipolar consistency, especially around high-motion segments and chunk boundaries.

6.7. Qualitative Analysis

Qualitative comparisons will be added around high-motion regions, chunk boundaries, and foreground-background separation. The planned figure will compare ground truth, original SEVA with uniform sampling, original SEVA with pose-adaptive sampling, original-loss fine-tuning, and epipolar-loss fine-tuning. Based on the quantitative ablations, pose-adaptive sampling is expected to be most visible near difficult camera-motion segments, while low-noise epipolar fine-tuning should help where source-target overlap provides a meaningful geometric signal.

7. Discussion and Limitations

Sampling versus fine-tuning. The current results suggest that sampling design is a strong practical lever under limited compute. Pose-adaptive sampling improves perceptual quality using the original SEVA weights, while fine-tuning requires additional memory and time.

Timestep-aware geometry. Epipolar loss does not help when applied across the uncontrolled full timestep range, but controlled low-noise ablations are more favorable. This is consistent with the fact that epipolar supervision is applied to decoded predicted clean images, which are less reliable at high-noise timesteps.

Need for geometric metrics. PSNR, SSIM, and LPIPS measure image-level similarity but do not fully capture whether the synthesized trajectory respects the requested cameras. Future evaluation should include epipolar consistency metrics, Sampson distance, thresholded symmetric epipolar distance, or multi-view consistency metrics such as MEt3R [1].

Limitations. The current quantitative results are from a single scene and should be extended to a multi-scene test set. The epipolar loss uses simple RGB+Sobel descriptors, which may be less robust than learned features. Pose-adaptive sampling may increase the number of anchor frames and therefore inference time. Finally, all conclusions are constrained by limited fine-tuning compute.

8. Conclusion

This project studied compute-efficient adaptation of SEVA for trajectory novel view synthesis. After correcting camera metadata and evaluation setup, the experiments compared diffusion-only fine-tuning, visibility-gated epipolar fine-tuning, and pose-adaptive sampling. The preliminary results show that full-timestep epipolar fine-tuning is not a controlled improvement over original-loss fine-tuning, but low-noise epipolar supervision improves over corresponding low-noise baselines. The results also show that pose-adaptive anchor sampling improves perceptual quality without any model update. These findings suggest that timestep-aware geometric supervision and camera-motion-aware sampling are promising directions for adapting large pretrained NVS diffusion models under limited compute.

References

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